

Assignment_4_cwood19

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Use k-Means for clustering on financial data gathered on 21 firms in the Pharmaceuticals.csv.

The following variables are recorded:

1. Market capitalization (in billions of dollars): Market_Cap,
2. Beta: Beta,
3. Price/earnings ratio: PE_Ratio,
4. Return on equity: ROE,
5. Return on assets: ROA,
6. Asset turnover: Asset_Turnover,
7. Leverage: Leverage,
8. Estimated revenue growth: Rev_Growth,
9. Net profit margin: Net_Profit_Margin,
10. Median recommendation (across major brokerages): Median_Recommendation,
11. Location of firm's headquarters: Location,
12. Stock exchange on which the firm is listed: Exchange

```
# Load packages
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
# Load dataset
rx <- read.csv("Pharmaceuticals.csv")

# Remove unused variables
rx <- rx[, !(names(rx) %in% c("Symbol", "Name"))]
```

A. Use only the numerical variables (1 to 9) to cluster the 21 firms. Justify the various choices made in conducting the cluster analysis, such as weights for different variables, the specific clustering algorithm(s) used, the number of clusters formed, and so on.

```
# Numerical variables (1-9)
rx_num <- rx[, c("Market_Cap",
                 "Beta",
                 "PE_Ratio",
                 "ROE",
                 "ROA",
                 "Asset_Turnover",
                 "Leverage",
```

```

        "Rev_Growth",
        "Net_Profit_Margin"]])

# Scale variables to avoid any dominating the clustering
rx_scaled <- scale(rx_num)

# Set seed to produce standardized results
set.seed(123)

# Utilize Elbow Method to determine number of clusters
em <- numeric(10)

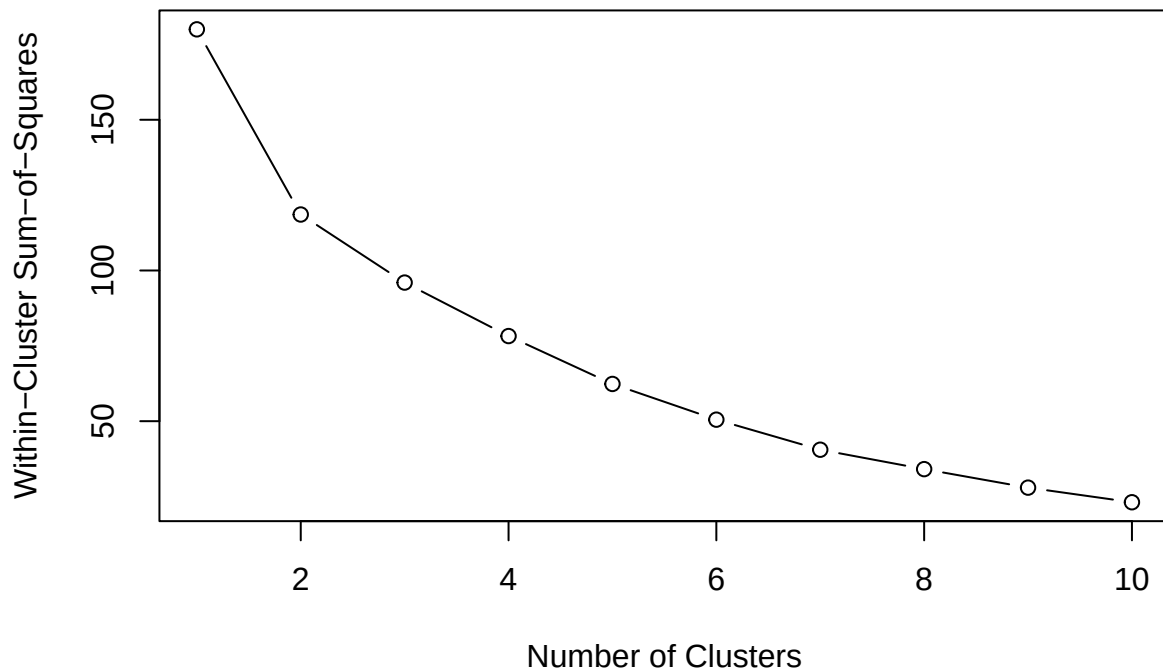
for(i in 1:10){
  em[i] <- kmeans(rx_scaled,
                  centers = i,
                  nstart = 25)$tot.withinss
}

# View em plot values
em

## [1] 180.00000 118.56934 95.99420 78.24600 62.35443 50.50985 40.53936
## [8] 34.09583 27.95526 23.11114

# Plot em
plot(1:10, em,
     type = "b",
     xlab = "Number of Clusters",
     ylab = "Within-Cluster Sum-of-Squares")

```



```
# fit the k-means model (ex: 3 clusters)
set.seed(123)
```

```
km <- kmeans(rx_scaled,
             centers = 3,
             nstart = 25)
```

```
km
```

```
## K-means clustering with 3 clusters of sizes 4, 11, 6
```

```
##
```

```
## Cluster means:
```

```
##   Market_Cap      Beta  PE_Ratio      ROE      ROA Asset_Turnover
## 1 -0.6125361  0.2698666  1.3143935 -0.9609057 -1.0174553   0.2306328
## 2  0.6733825 -0.3586419 -0.2763512  0.6565978  0.8344159   0.4612656
## 3 -0.8261772  0.4775991 -0.3696184 -0.5631589 -0.8514589  -0.9994088
```

```
##   Leverage Rev_Growth Net_Profit_Margin
```

```
## 1 -0.3592866 -0.5757385      -1.3784169
## 2 -0.3331068 -0.2902163       0.6823310
## 3  0.8502201  0.9158889      -0.3319956
```

```
##
```

```
## Clustering vector:
```

```
## [1] 2 1 1 2 3 1 2 3 3 2 2 3 2 3 2 2 1 2 3 2
```

```
##
```

```
## Within cluster sum of squares by cluster:
```

```
## [1] 20.54199 43.30886 32.14336
```

```
## (between_SS / total_SS = 46.7 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
##
# Label clusters
rx$Cluster <- factor(km$cluster)
# Examine clusters
table(rx$Cluster)

##
##  1  2  3
##  4 11  6
```

A) The nine numeric variables utilized measured on different scales (Market cap in billions, Beta < 1, etc.) are standardized / scaled for clustering. K-means clustering has the capacity for grouping observations based on their numerical characteristics. The elbow method is used to evaluate the number of clusters to select (based on diminishing returns after $k = 3$) and interprets the use of cluster size 4, 11, and 6 firms.

B. Interpret the clusters with respect to the numerical variables used in forming the clusters.

```
# Calculate cluster centroids
aggregate(rx_num,
          by = list(Cluster = rx$Cluster),
          mean)
```

	Cluster	Market_Cap	Beta	PE_Ratio	ROE	ROA	Asset_Turnover	Leverage
##	1	21.75500	0.5950000	46.90000	11.3	5.100000	0.7500000	0.3050000
##	2	97.11364	0.4336364	20.95455	35.7	14.954545	0.8000000	0.3254545
##	3	9.23500	0.6483333	19.43333	17.3	5.983333	0.4833333	1.2500000

	Cluster	Rev_Growth	Net_Profit_Margin
##	1	7.01000	6.65000
##	2	10.16455	20.17273
##	3	23.49000	13.51667

B) Cluster 1 reveals the largest price to earnings ratio in spite of the low market cap, ROE, and ROA. Cluster 2 boasts the largest market cap and lowest beta; likely an established industry leader. Cluster 3 yields the highest revenue growth potential with its high risk.

C. Is there a pattern in the clusters with respect to the numerical variables (10 to 12)? (those not used in forming the clusters)

```
# Generate clusters to view tables based on variable 10, 11, 12
table(rx$Cluster, rx$Median_Recommendation)
```

	Cluster	Hold	Moderate	Buy	Moderate	Sell	Strong	Buy
##	1	2	1	0	1			
##	2	6	3	2	0			
##	3	1	3	2	0			

```
table(rx$Cluster, rx$Location)
```

```
##
##      CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
## 1      1      0      1      0      0      1  1
## 2      0      0      0      0      0      2  8
## 3      0      1      0      1      0      0  4
```

```
table(rx$Cluster, rx$Exchange)
```

```
##
##      AMEX NASDAQ NYSE
## 1      0      0    4
## 2      0      0   11
## 3      1      1    4
```

C-Recommendation) Cluster 1 is the only cluster to show a strong buy while Cluster 2 & 3 are pushing a moderate sell. Even though Cluster 2 maintains the most valuable firms for market cap, there is a high potential with Cluster 1.

C-Location) Cluster 2 suggests that the largest and most profitable sampled companies are based in the US. Since cluster 3 is pushing to sell and also has the majority based in the US, it may reveal the US having a higher risk-reward market.

C-Exchange) The majority of firms exchange in NYSE regardless of cluster, but those that do not all belong to cluster 3 with high leverage and growth.

D. Provide an appropriate name for each cluster using any or all of the variables in the dataset.

```
# Show 3 clusters aggregate data
cluster_summary <- aggregate(rx_num,
                             by = list(Cluster = rx$Cluster),
                             mean)

cluster_summary[, -1] <- round(cluster_summary[, -1], 2)

cluster_summary
```

```
## Cluster Market_Cap Beta PE_Ratio ROE ROA Asset_Turnover Leverage
## 1      1      21.75 0.60  46.90 11.3  5.10      0.75    0.30
## 2      2      97.11 0.43  20.95 35.7 14.95      0.80    0.33
## 3      3       9.23 0.65  19.43 17.3  5.98      0.48    1.25
## Rev_Growth Net_Profit_Margin
## 1      7.01      6.65
## 2     10.16     20.17
## 3     23.49     13.52
```

D-Cluster-1) High Value, Small Cap

D-Cluster-2) Stable High Cap

D-Cluster-3) Growth Factor